

DATE: 22/2/26

DAY: ___/___/___

Assignment 01

Analysis of Algorithms

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Q1

Smallest of 3 numbers:-

Algorithm: Find Smallest (A, B, C)

{ The algorithm is used to read three numbers and Print the Smallest. }

Step - 1 [Read three numbers]

Read (A, B, C)

cin >> A >> B >> C;

Step - 2 [check A]

If (A < B && A < C)

Print A is smallest.

Step - 3 [check B]

If (B < A && B < C)

Print B is smallest.

Step - 4 [Else]

c is smallest

Step - 5 [Exit]

Exit with Return 0.

C++ :

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {  
    int A, B, C;
```

```
    cout << "Enter 3 numbers:";
```

```
    cin >> A >> B >> C;
```

```
    if (A <= B && A <= C) {
```

```
        cout << A << " is smallest" << endl;
```

```
    }
```

```
    else if (B <= A && B <= C) {
```

```
        cout << B << " is smallest" << endl;
```

```
    }
```

```
    else {
```

```
        cout << C << " is smallest" <<  
        endl;
```

```
    return 0;
```

```
}
```


~~Ex (2)~~

Count Even in Array :

Algorithm CountEven [Arr[], Cout]
{ used to read an array
of 10 integer and count the
even number }

Step - 1 [Declare an array]
arr [10]

Step - 2 [initialize Count]
Cout = 0

Step - 3 [Run loop]
For (i = 0 to 9)
Read Arr[i]

Step - 4 [check condition]
If Arr[i] MOD 2 == 0
Then Count = Count + 1
End If.

Step - 5 [Print Count]
cout << Count ;

Step - 6 [Exit]
Exit with return

C++:-

```

#include <iostream>
using namespace std;
int main () {
    int arr[10];
    int count = 0;
    cout << "Enter 10 Integer << endl;

    for (i = 0; i < 10; i++) {
        cin >> arr[i]

        if (arr[i] % 2 == 0) {
            count++;
        }
    }

    cout << "Total even << count << endl;

    return 0;
}

```

* ————— *

3

Product using sub-Algo.

Algorithm Multiply (x, y) (num1, num2).

§ { Read two values and find their Product using a Sub - algorithm }

Step - 1 [Create sub- Algo]
Multiply (int x , int y)
return $x * y$;

Step - 2 [Read two numbers]
Read (num1 , num2)

Step - 3 [Call the sub- Algo]
Call (num1 , num2)

Step - 4 [Print result]
cout << result << endl;

Step - 5 : [Exit]
Exit with return.

§. ————— §.

C++ :

```
#Include <iostream>
using namespace std;
int multiply (int x , int y)
```



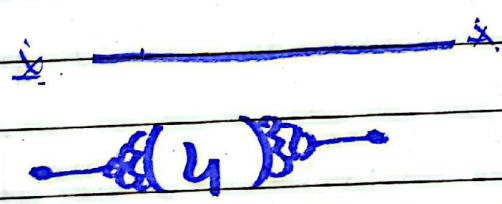
```

{
    return x*y;
}

int main () {
    int num1, num2, result;
    cout << "Enter two values ";
    cin >> num1, num2;

    result = Multiply (num1, num2);
    cout << "The product is"
    << result << endl;
    return 0;
}

```



Sum of two 3x3 Matrices.

Algorithm MatrixAdd (A,B)
 { This algorithm used
 to find the sum of two
 3x3 matrices. }

Step -1 [Declare matrices]
 A[3][3] B[3][3], C[3][3]

Step -2 [use loop]

For $i = 0$ to 2

For $j = 0$ to 2

$C[i][j] = A[i][j] + B[i][j]$

Step - 3 [End loop & Print]

End For

`cout << C[i][j] << " ";`

Step - 4 [Exit]

Exit with return

C++ :

Include <iostream>

using namespace std;

int main () {

int A [3][3], B [3][3], C [3][3];

`cout << "Enter 9 elements ;`

`for (int i = 0 ; i < 3 ; i++) {`

`for (int j = 0 ; j < 3 ; j++) {`

`cin >> A[i][j];`

`}`
`}`

`cout << "Sum of 2 Matrices ;`

`for (int i = 0 ; i < 3 ; i++) {`

`for (int j = 0 ; j < 3 ; j++) {`

`C[i][j] = A[i][j] + B[i][j]`

`cout << C[i][j] << " ";`


```

    }
    cout << endl;
}
return 0;
}

```

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Prime number check:-

Algorithm CheckPrime (N : I)
 { Algorithm used to read
 number N and print wheth
 er its Prime or Not. }

Step - 1 [Read N]
 Read (N)

Step - 2 [check if]
 If $N \neq 1$ Then
 Print "Not Prime."
 Exit

Step - 3 [check calculation]
 For $i = 2$ to $(N/2)$
 If $N \text{ MOD } i == 0$ Then
 Print "Not Prime"
 Exit

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Step - 4 [End loop]
End If & Print "is Prime"

Step - 5 [Exit]
Exit with return.

C++:-

```
#include <iostream>
using namespace std;
int main () {

    int N;
    bool isPrime = true;

    cout << "Enter a number N:";
    cin >> N;
    if (N <= 1) {

        isPrime = false;
    } else {

        for (int i = 2; i <= N / 2; i++) {

            if (N % i == 0) {

                isPrime = false;

                break;
            }
        }
    }
}
```



```
if (isPrime) {  
    cout << N << " is Prime << endl;  
}  
else {  
    cout << N << " is not  
    Prime " << endl;  
}  
return 0;  
}
```

—————